

Event-Based Camera Questions – Lab 2

Pravin Oli, u1999699, pravin.oli.08@gmail.com, IFROS

2025 March 24

Part 1: Questions and Answers

1. What could cause an event to have a positive or negative polarity?

The increase in brightness and decrease in brightness cause an event to have +ve and -ve polarity because Event-based cameras detect brightness changes at individual pixels and generate events accordingly. Each event is labeled with a polarity, indicating whether the brightness increased (positive) or decreased (negative).
For example:

- **Positive Polarity (brightness increases):** When an object moves into a brighter region or light source turns on, or bright object moves across a dark surface.
- **Negative Polarity (brightness decreases):** When a bright object blocks a darker background, or dark object crosses a light surface or light source dims or turns off.

2. Given the timestamp resolution, calculate the maximum theoretical event rate this sensor can handle. How would this impact applications like high-speed object tracking? Explain.

The **DAVIS346B** sensor has a resolution of $260 \times 346 = 89,960$ pixels, with each pixel capable of registering an event every microsecond (1M events/second).

$$\begin{aligned}\text{Max events per second} &= \text{Number of pixels} \times \text{Max events per pixel per second} \\ &= 89,960 \times 1,000,000 \\ &= 89.96 \times 10^9 \text{ events/second}\end{aligned}$$

Applications: This extremely high event rate supports real-time applications for example:

- Monitoring rotating turbines or propellers and also event based vibration analysis.
- Capturing fluid flow dynamics.
- Tracking high-speed objects (e.g., drones, robotic arms)
- Application in reducing motion blur due to frame based camera.

3. How to identify and filter out noise without significantly impacting true events?

Noise events captured by EBC is due to thermal drift or flickering lights and these noise are usually random and unstructured. We can filter these noises by following strategies:

- **Optical flow filtering:** Detect and retain events that follow coherent motion patterns.
- **Time-based filtering:** Remove isolated events with no nearby temporal neighbors.
- **Intensity thresholding:** Ignore events triggered by very small brightness fluctuations.
- **Spatial filtering:** Eliminate events lacking adjacent events within a spatial neighborhood.

Part 2: Questions and Answers

1. How does the choice of time window affect the visibility of objects?

The time window defines how long the events are accumulated to form an image. Its length greatly influences image sharpness and motion clarity. Shorter windows are preferred when clarity and precision are required, especially for slow or detailed motion.

For Example:

- **Short windows** (e.g., 0.2s): Capture fewer events, resulting in cleaner and sharper object edges with less blur.
- **Long windows** (e.g., 1.0s): Accumulate more events, causing motion blur and thicker edges due to overlapping events.

2. Would this representation work well for detecting fast-moving objects? Why or why not?

Yes, this representation is effective for tracking fast motion when the time window is small enough to capture clean, non-overlapping event bursts.

Choosing the appropriate short time window balances image clarity and temporal context. For high-speed motion tracking, shorter windows offer greater accuracy.

- **Short time windows** isolate fast motion and preserve shape and trajectory.
- **Long time windows** cause overlapping and blurred motion paths, making it harder to distinguish separate movements.

We can clearly see the difference from the result of this lab:

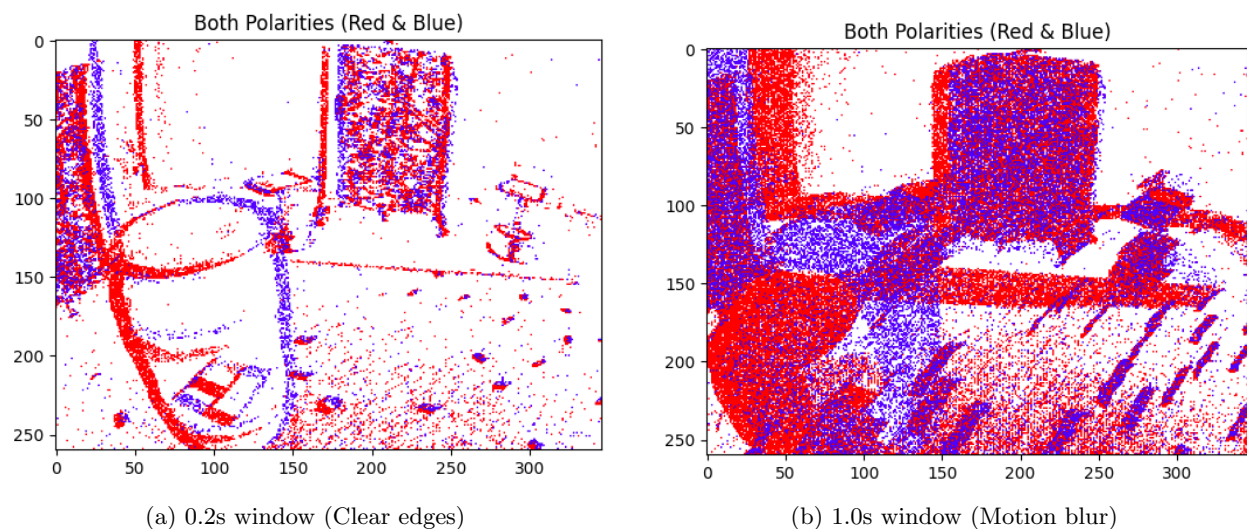


Figure 1: Effect of different time windows on event-based image clarity.

Part 3: Comparative Event Image Analysis (0.2s vs 1.0s)

1. How clear is the obtained events image? Why?

The clarity of the event image depends strongly on the accumulation period. With a shorter time window of 0.2 seconds, the image retains sharp boundaries and fine object contours. This is because fewer events are overlaid, preserving spatial details.

In contrast, the 1.0-second image appears more cluttered. The prolonged event accumulation introduces overlapping motion patterns, resulting in thicker edges and reduced clarity—especially when objects are in motion.

2. How do the images change after applying the frame-based approach? Why?

With the frame-based approach, instead of merging all events into one image, we split the event stream into several shorter frames.

Frame-wise sequence for 0.2s:

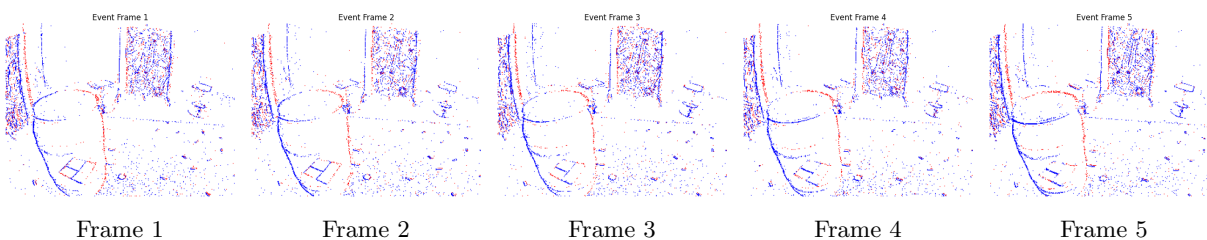


Figure 2: Frame-based representation for 0.2s sequence (sharper, less overlap)

Frame-wise sequence for 1.0s:

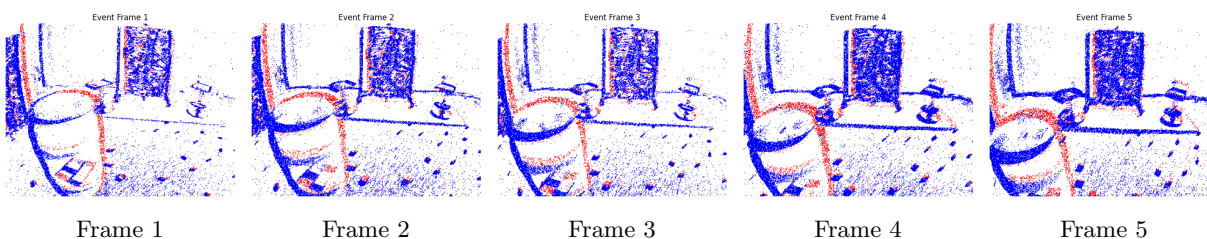


Figure 3: Frame-based representation for 1.0s sequence (noisier, motion overlap)

- For the **0.2s sequence**, the frames are clean and show distinct, smooth transitions in object motion. This allows for easier tracking of movement over time.
- For the **1.0s sequence**, although more movement is captured, the frames become more cluttered with noise and overlapping event trails, making individual object paths harder to isolate.

Thus, frame-based encoding helps break down motion temporally, making it easier to observe dynamic behavior, though it still suffers from blur in long windows.

General Questions (Applicable to Both Cases)

1. How would the clarity of the events image change if we increase the camera velocity for the same scene?

If the camera moves faster while observing the same scene, more rapid intensity changes will be detected, producing a higher number of events within the same time window. This leads to:

- Increased event density
- Higher chances of overlapping motion trails
- Greater motion blur, especially with longer time windows

Therefore, clarity decreases as camera velocity increases, particularly when using a large accumulation window.

2. Which event representation is more informative: one-image or frame-based approach? Why?

The **frame-based approach** is more informative. Unlike a single image that compresses all events into one snapshot—losing motion sequence and direction—frame-based encoding captures the evolution of motion over time. This makes it easier to analyze:

- Temporal order of events
- Object trajectories
- Direction and speed of movement

3. What information is lost with simple event count encoding? Suggest alternatives.

Frame-based images make temporal dynamics visible, allowing better understanding of motion. The 0.2s version shows finer transitions, while the 1.0s version accumulates more noise and overlap. Information lost with simple event count encoding are :

- **Timestamps:** Exact timing of events is lost, preventing temporal reasoning.
- **Event order:** We cannot tell in which order events occurred.
- **Motion direction:** There's no indication of whether an object is moving left, right, etc.

Alternative Representations:

- **Time Surfaces:** Store the timestamp of the most recent event at each pixel.
- **Event Flow (Optical Flow):** Estimate direction and speed of pixel-level motion.
- **Spiking Neural Networks (SNNs):** Handle asynchronous events more naturally for motion analysis.